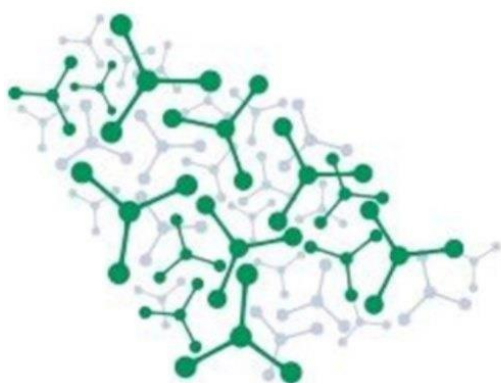


Last name _____

56th International Chemistry Olympiad



56th IChO International
Chemistry Olympiad
Saudi Arabia 2024

Qualifying theoretical round

Faculty of Chemistry, Lomonosov Moscow State
University (MSU)
16 June 2024

Last name _____

General Instructions

- This theoretical round problem set contains **9 problems** with answer sheets and spans **45 pages**.
- The maximum possible score is 60.
- You have 5 hours to complete the problems
- Begin working after the "Start" command.
- Wherever required, support your answers with calculations.
- Only the sheets on which you are working, pens, a ruler, a calculator, water, and chocolate may be present at your workstation. All electronic devices are prohibited.
- **Write your solutions only on the answer sheets. The back side of the problem sheets may be used as scratch paper. Scratch work will not be graded.**
- You must stop working immediately after the "Stop" command is given. If you do not stop writing within 1 minute, your result for the entire olympiad will be annulled.

We wish you good luck!

Last name _____

Periodic Table

1 H 1.008																	18 He 4.003
3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.30											13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.06	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.63	33 As 74.92	34 Se 78.97	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.95	43 Tc -	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3
55 Cs 132.9	56 Ba 137.3	57-71	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po -	85 At -	86 Rn -
87 Fr -	88 Ra -	89-103	104 Rf -	105 Db -	106 Sg -	107 Bh -	108 Hs -	109 Mt -	110 Ds -	111 Rg -	112 Cn -	113 Nh -	114 Fl -	115 Mc -	116 Lv -	117 Ts -	118 Og -

57 La 138.9	58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm -	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
89 Ac -	90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np -	94 Pu -	95 Am -	96 Cm -	97 Bk -	98 Cf -	99 Es -	100 Fm -	101 Md -	102 No -	103 Lr -

Last name _____

Физические константы, формулы и уравнения

Постоянная Авогадро $N_A = 6.022 \times 10^{23} \text{ моль}^{-1}$

Универсальная газовая постоянная $R = 8.314 \text{ Дж} \cdot \text{К}^{-1} \cdot \text{моль}^{-1}$

Постоянная Фарадея $F = 9.6485 \times 10^4 \text{ Кл} \cdot \text{моль}^{-1}$

Стандартное давление $p^\circ = 1 \text{ бар} = 10^5 \text{ Па}$

Ноль по шкале Цельсия 273.15 К

Атомная единица массы $1 \text{ а.е.м.} = 1.6605 \times 10^{-27} \text{ кг}$

Ангстрем $1 \text{ \AA} = 10^{-10} \text{ м}$

Ионное произведение воды $K_w = 1.00 \times 10^{-14}$

Энергия Гиббса $G = H - TS$

Связь константы равновесия с энергией Гиббса $K_p = \exp\left(-\frac{\Delta G^\circ}{RT}\right)$

Связь K_p и K_c $K_p = K_c \left(\frac{C^\circ RT}{p^\circ}\right)^{\Delta \nu}$

Зависимость концентрации от времени
0-й порядок: $C(t) = C(0) - kt$
1-й порядок: $\ln C(t) = \ln C(0) - kt$
2-й порядок: $1/C(t) = 1/C(0) + kt$

Уравнение Гендерсона-Хассельбальха $\text{pH} = \text{p}K_a + \lg \frac{[\text{A}^-]}{[\text{HA}]}$

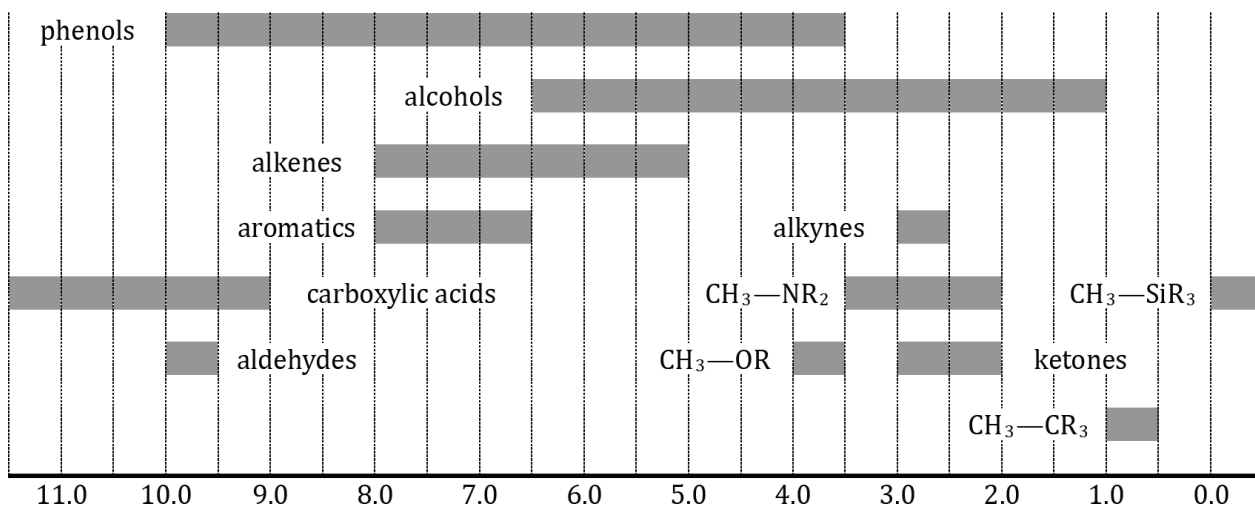
Уравнение Аррениуса $k = Ae^{-\frac{E_a}{RT}}$

Last name _____

NMR Spectroscopy

^1H NMR

Chemical shifts of hydrogen (in ppm /TMS)



Spin-spin coupling constants

Structural Type	J (Hz)
$\text{H}-\text{C}-(\text{C})_n-\text{C}-\text{H}$	0 (unless in a rigid ideal orientation)
$\text{H}_3\text{C}-\text{CH}_2-\text{X}$	6 to 8
$\begin{array}{c} \text{H}_3\text{C} \\ \\ \text{H}_3\text{C}-\text{CH}-\text{X} \end{array}$	5 to 7
$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{C}-\text{H} \\ \quad \\ \text{X} \quad \text{Y} \end{array}$	2 to 12 (depends on dihedral angle and the nature of X and Y)
$\begin{array}{c} \text{O} \\ \\ \text{H}-\text{C}-\text{C}-\text{H} \\ \quad \\ \text{H} \quad \text{H} \end{array}$	0.5 to 3
$\begin{array}{c} \text{H} \\ \\ \text{C} \\ \\ \text{H} \end{array}$	12 to 15 (must be diastereotopic)

Structural Type	J (Hz)
$\begin{array}{c} \text{H} \\ \\ \text{C}=\text{C} \\ \quad \\ \text{H} \end{array}$	12 to 18
$\begin{array}{c} \text{H} \quad \text{H} \\ \diagdown \quad \diagup \\ \text{C}=\text{C} \end{array}$	7 to 12
$\begin{array}{c} \text{H} \\ \\ \text{C}=\text{C} \\ \quad \\ \text{H} \end{array}$	0.5 to 3
$\begin{array}{c} \text{H} \\ \\ \text{C}=\text{C} \\ \quad \\ \text{C}-\text{H} \end{array}$	3 to 11 (depends on dihedral angle)
$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{C}\equiv\text{C}-\text{H} \\ \\ \text{H} \end{array}$	2 to 3
	o 6 to 9 m 1 to 3 p 0 to 1

Last name _____

Problem List

Name	Points
1. Orbital Inversion	7
2. Decomposition Equilibrium Under Various Conditions	6
3. Russian analog of Prussian Blue	5
4. Total Synthesis of a Macrolactone	7
5. Acid-Base Titration	7
6. Kinetics of Oxidation of Alcohols and Thiols by Dichromate	7
7. "Guardians" and conductors	7
8. Polymers	7
9. Three-Stage Scheme of Enzymatic Reactions	7
Total – 60	

Last name _____

Problem 1. Orbital Inversion (7 points)

Questions	1	2	3	4	5	6	7	Total
Points	2	1.5	1	1	2	1.5	1	10
Score								

The relative energy ordering of molecular orbitals determines the sequence in which they are filled with electrons. In some cases, the expected orbital ordering may be disrupted, which significantly affects the electron distribution in the molecule and its chemical properties. Orbital inversion is characteristic of diatomic molecules in the first half of the 2nd period, due to the interaction of 2s and 2p orbitals that are close in energy and symmetry.

1. Draw the molecular orbital diagram for the homonuclear diatomic molecule B_2 . Label the symmetry of each molecular orbital, indicate the antibonding orbitals, fill the diagram with electrons, and determine the expected bond order in the molecule.

2. In some borides, discrete B_2^{2-} anions can be found. Compare the bond length and the first ionization energy in the dianion and in the natural neutral boron molecule (B_2)

Last name _____

Relatively recently, the inversion of metal and ligand orbitals has been studied for transition metal complexes. The main reason for the change in the order of orbitals in this case is the relatively high energy of the ligand frontier orbitals compared to the metal frontier orbitals.

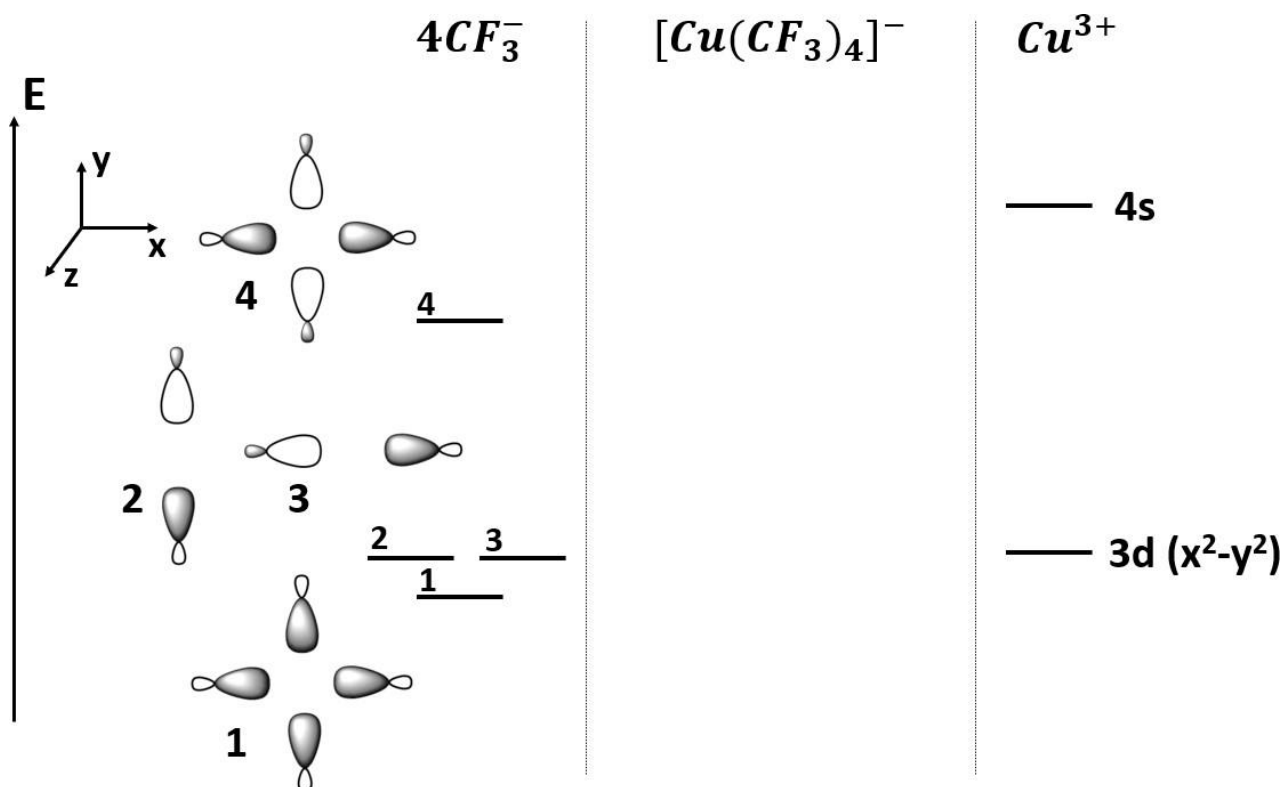
In the work of R. Hoffmann and co-authors (*Chemical Reviews* 2016), a search for examples of orbital inversion was conducted using quantum chemical calculations. One of the research objects was a model chromium(0) sigma-complex with hydride anions, for which the energy of the HOMO was artificially raised from -13.6 eV to -8.6 eV.

3. Suggest how the electron density on chromium should change (increase, decrease, practically does not change) with an increase in the energy of the hydrogen orbital..

One of the first experimentally recorded cases of inversion was observed in the square planar anion $[\text{Cu}(\text{CF}_3)_4]^-$, which was obtained by the oxidation of CuCF_3 . The figure below shows some frontier orbitals of the ligands and the metal. The remaining *d*-orbitals of copper have lower energy.

4. Which of the provided ligand group orbitals and metal orbitals can interact effectively? Depict the corresponding combinations..

Last name _____



5. In the diagram provided above:

- Fill the ligand and metal orbitals with electrons, based on the expected structure of the complex.
- In the middle part of the drawing, schematically represent the orbital diagram of the complex, indicating the original "parent" orbitals with dashed lines. Fill it with electrons.

6. Based on the filled-in diagram, estimate:

- The average Cu-C bond order in the complex.
- The formal oxidation state of copper (use only integer oxidation states).

Last name _____

7. In addition to the orbitals shown, the metal also has four $3d$ and three $4p$ orbitals. Which of these orbitals could interact effectively with the ligand group orbitals? Represent these interactions schematically and also indicate whether they lead to a strengthening of bonding in $[\text{Cu}(\text{CF}_3)_4]^-$.

Last name _____

Problem 2. Decomposition Equilibrium Under Various Conditions

Question	1	2	3	4	5	Total
Points	5	4	4	2	5	20
Score						

Upon heating, gaseous nitrogen dioxide reversibly decomposes into two substances, one of which is an element in its standard state. At a temperature of 450 °C and a total pressure of 1 bar, the degree of decomposition is 58.1%.

1. What are the equilibrium constants K_p and K_c for the reaction with the smallest integer coefficients at this temperature?

Calculation

$K_p =$ _____

$K_c =$ _____

Last name _____

0.100 mol of NO_2 was introduced into an evacuated reactor with a volume of 1.00 L, heated to 450 °C and held at that temperature until the pressure stopped changing.

2. What percentage of NO_2 decomposed under these conditions?

Calculation

$\alpha =$ _____

0.100 mol of NO_2 , heated to 450 °C, was introduced into an adiabatically isolated reactor with a volume of 1.00 L and held until the pressure stopped changing, reaching a final value of 6.63 bar. The degree of decomposition was found to be 30.0%.

2. Find the final temperature in the reactor and the enthalpy of the decomposition reaction of NO_2 .

Calculation

$T =$ _____K

Last name _____

$$\Delta_r H = \text{_____ kJ/mol}$$

3. Sketch qualitative graphs of K_p for the decomposition reactions as a function of temperature and total pressure. You may use any coordinate system convenient for you.

Last name _____

0.200 mol of NO_2 was introduced into an evacuated vessel with a volumen of 1.00 L containing 6.54 g of zinc poder, heated to 450 °C, and held at that temperature until the pressure stopped changing. The standard enthalpy of formation of ZnO is -350.6 kJ/mol.

4. a) Determine the enthalpy of the reaction between Zn and NO_2 . The dependence of enthalpy on pressure may be neglected.
b) Calculate the equilibrium pressure in the reactor.

Calc.

a)

$$\Delta_r H = \text{_____ kJ/mol}$$

b)

$$p = \text{_____ bar}$$

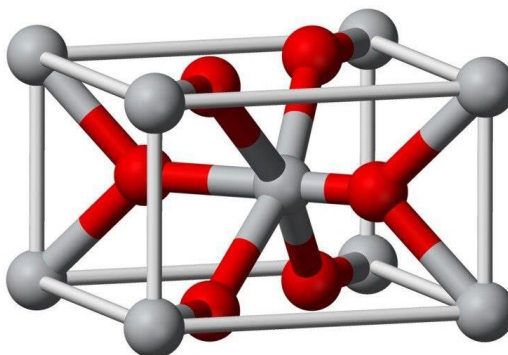
Last name _____

Problem 3. Russian Analog of Prussian Blue (5 points)

Question	1	2	3	4	5	6	Total
Points	1	1	1	4	2	1	10
Score							

A grey-black powder of substance X_1 exhibits ferromagnetic properties and has the structure shown in the figure. It contains 85,32% by mass of element Z.

1. Identify X_1 . Name its structure type after the most common mineral.



2. To obtain X_1 , colorless crystals of X_2 were heated at 400°C . When X_2 is treated with potassium hydroxide, a gas X_3 is released and colorless water-soluble crystals X_4 are formed, which are isostructural with potassium perchlorate. Identify the unknown substances and write the reaction equations

Last name _____

3. Yellow-orange water-soluble crystals of Y_1 , were obtained by Russian chemists via fusión in a quartz ampoule at 600°C of substance X_1 , potassium cyanide, and sulfur, with potassium cyanide taken in excess. The substance was purified by recrystallization from aqueous ethanol. Under the reaction conditions, potassium cyanide plays various roles:

- (1) Complex-forming agent
- (2) Ligand
- (3) Reducing Agent
- (4) Oxidising Agent
- (5) Reaction medium («solvent»)

Indicate the number of correct answers.

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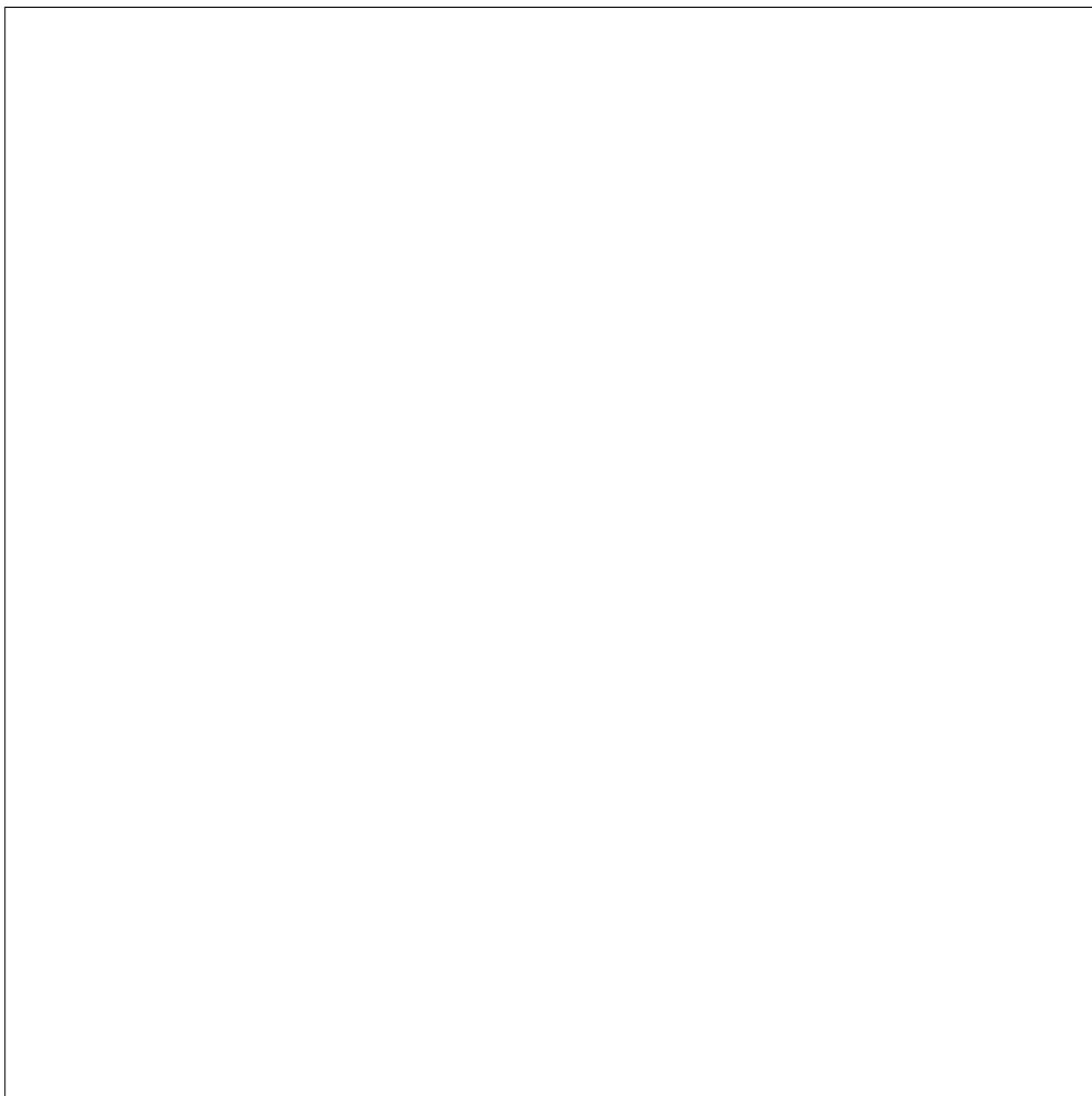
A 5052 mg sample of substance Y_1 was fused with potassium chlorate and potassium hydroxide. All of the potassium chlorate was consumed in the oxidation. During the reaction, 201.6 mL of gas (S.T.P.) was evolved, with a density slightly less than that of air and practically insoluble in water. The solid residue from the calcination dissolves completely in water; this solution, upon treatment with excess barium chloride solution, gives a precipitate, which upon treatment with excess sulfuric acid evolves 403.2 mL (S.T.P.) of gas, and the mass of the precipitate decreases to 5592 mg. From the solution obtained after addition of barium chloride and separation of the precipitate, a mixture of two salts can be isolated — X_4 (see part 2) with a mass of 5202 mg (this mass implies quantitative recovery of X_4) and a second salt isostructural with halite.

4. Write the empirical and true (see part 5) formula of substance Y_1 and the equations for its synthesis and oxidative fusion reactions. Confirm with calculations

Last name _____

Last name _____

5. The anion in Y_1 has a symmetric structure, with sulfur atoms located at the vertices of a cube. Draw the structure of this anion.



6. When an aqueous solution of Y_1 is treated with iron(III) nitrate solution, a black precipitate forms whose structure resembles the insoluble form of Prussian blue. Write the equation for the reaction.

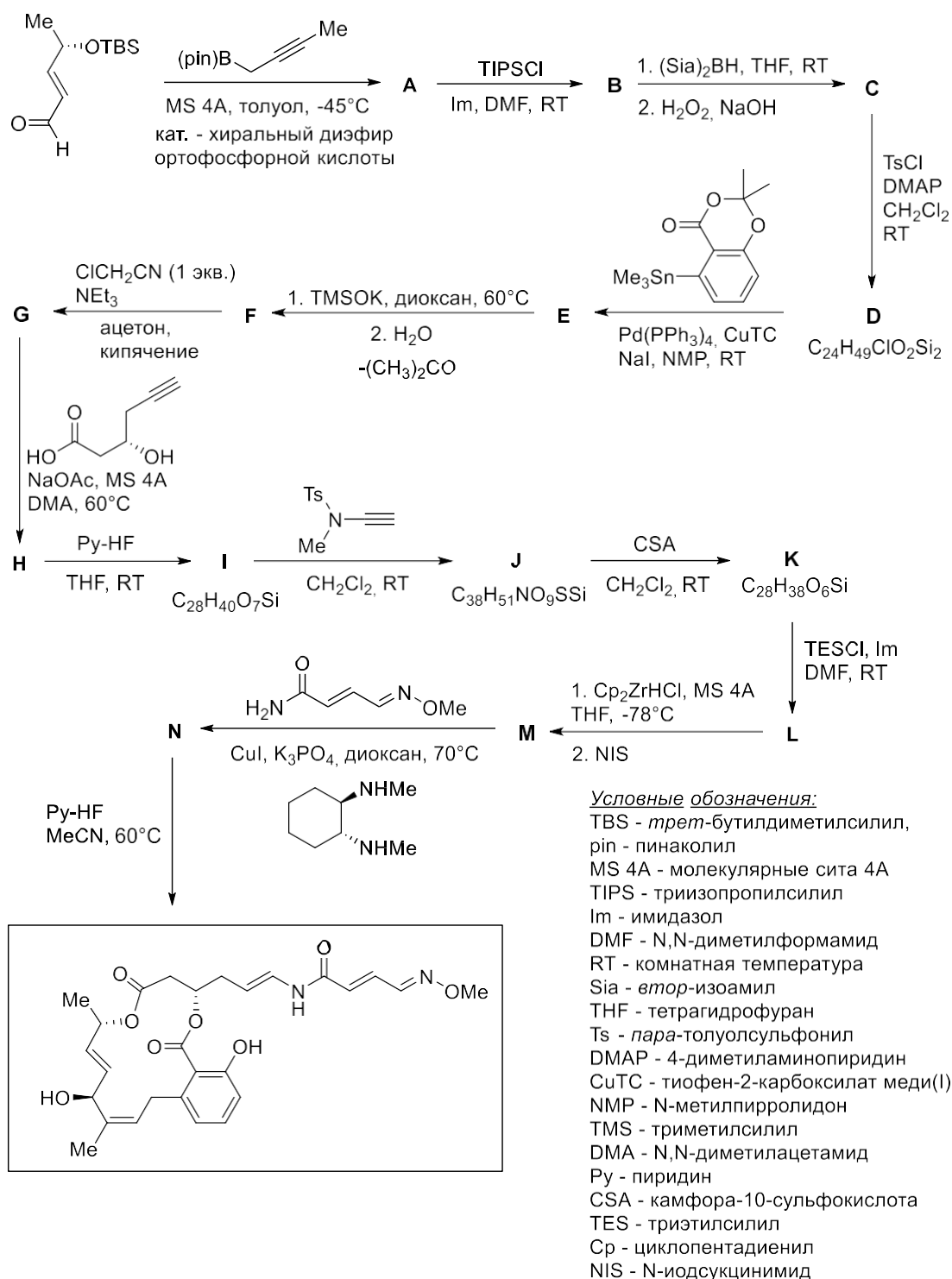


Last name _____

Problem 4. Total Synthesis of a Macrolactone (7 points)

Question	1	2	3	Total
Points	14	11	2	27
Score				

In 2024, a group of Japanese scientists published the total synthesis of a natural macrolactone that exhibits antitumor activity by inhibiting the V-ATPase enzyme. A scheme of this synthesis is provided below.



Last name _____

1. Provide the structural formulas for compounds **A–N**. It is not necessary to specify the configuration of stereocenters. It is additionally known that a transesterification reaction occurs during the transformation stage of **G** to **H**.

A	B
C	D
E	F
G	H

Last name _____

I	J
K	L
M	N

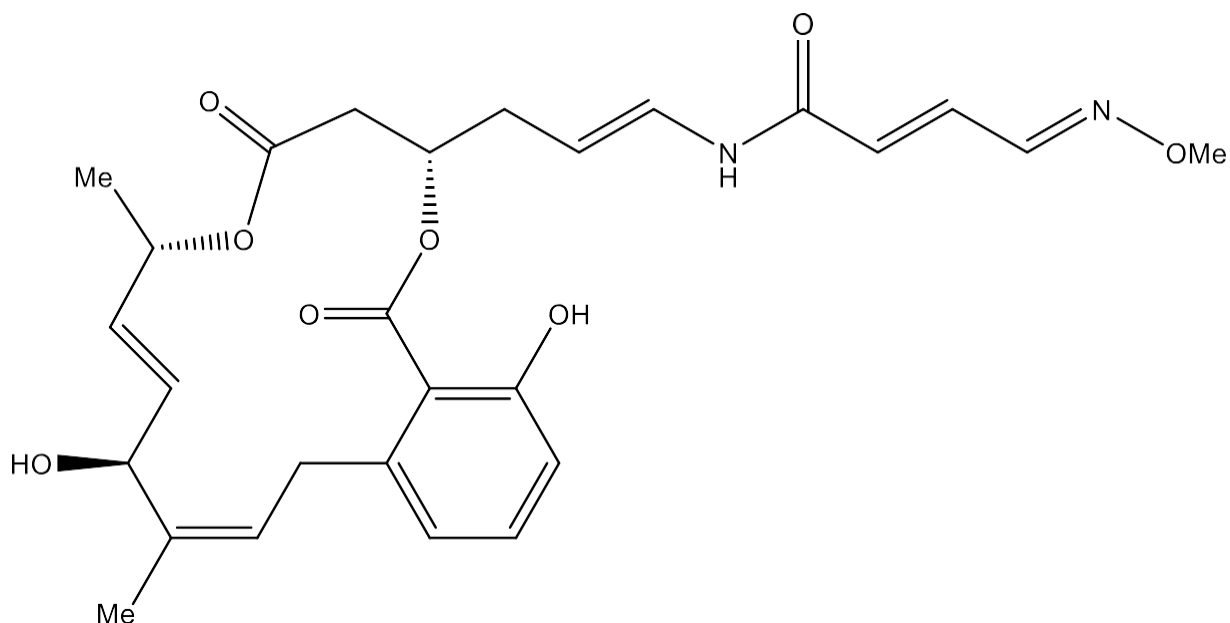
Last name _____

Below is a list of signals for the target product in the ^1H NMR, recorded in CD_3OD on a частотой 500 MHz instrument. Some of the signals are designated by numbers.

Number	Signal	Number	Signal
1	7.86 (d, $J = 10.3$ Hz, 1H)	8	5.23 (dq, $J = 8.6, 6.6$ Hz, 1H)
2	7.16 (dd, $J = 15.5, 10.3$ Hz, 1H)		5.18 (m, 1H)
	7.15 (dd, $J = 8.0, 7.5$ Hz, 1H)	9	4.79 (d, $J = 8.8$ Hz, 1H)
3	6.87 (d, $J = 14.0$ Hz, 1H)		3.91 (s, 3H)
	6.69 (d, $J = 8.0$ Hz, 1H)		3.22 (dd, $J = 17.5, 8.6$ Hz, 1H)
	6.64 (d, $J = 7.5$, 1H)		2.95 (brd, $J = 17.5$ Hz, 1H)
4	6.29 (d, $J = 15.5$ Hz, 1H)		2.70 (dd, $J = 16.6, 2.0$ Hz, 1H)
5	5.68 (dd, $J = 15.2, 8.8$ Hz, 1H)		2.60 (dd, $J = 16.6, 10.6$ Hz, 1H)
	5.61 (dtd, $J = 10.6, 5.4, 2.0$ Hz, 1H)		2.50 (m, 2H)
6	5.50 (dd, $J = 15.2, 8.6$ Hz, 1H)	10	1.80 (s, 3H)
7	5.37 (dt, $J = 14.0, 7.7$ Hz, 1H)	11	1.35 (d, $J = 6.6$ Hz, 3H)

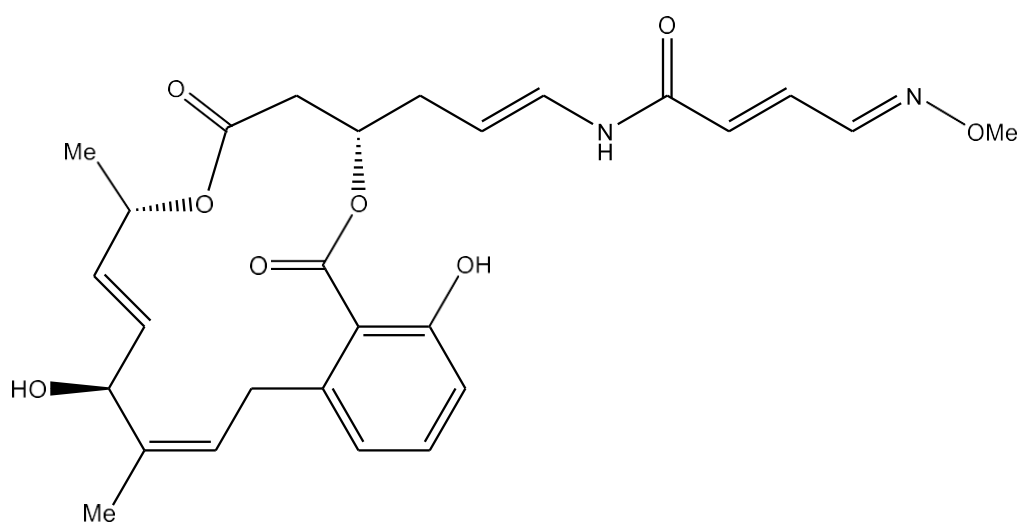
Designations: d – doublet, dd – doublet of doublets, dtd – doublet of triplets of doublets, dt – doublet of triplets, dq – doublet of quartets, m – multiplet, s – singlet, brd – broad doublet, J – spin-spin coupling constant.

- Label these numbers on the protons corresponding to the designated signals in the structural formula of the product:



Last name _____

3. Why do some protons of the target product not appear in the ^1H NMR? Mark these protons on the structural formula with the symbol "X".



Last name _____

Problem 5. Acid-Base Titrations (7 points)

Question	1	2	3	4	5	6	Total
Points	1.5	3	3	9	1.5	2	20
Score							

A young chemist once found a jar in the laboratory labeled PCl_5 . Inside were yellowish crystals and a transparent liquid. When the young chemist opened the jar, he noticed an unpleasant, pungent smell and concluded that the PCl_5 had partially hydrolyzed.

1. Write the reaction equation proposed by the young chemist.

To test his hypothesis and determine how extensively the sample had hydrolyzed, the young chemist decided to carry out an acid-base titration. To do this, he dissolved a small amount of the sample from the jar in excess water. To confirm that the reaction had gone to completion, the chemist measured the pH of the resulting solution.

2. Calculate the pH of the solution upon complete dissolution of 5 mmol of PCl_5 in 1 L of water. Assume that all HCl remains in solution and that the volume of the solution does not change appreciably. For phosphoric acid: $\text{p}K_{\text{a}1} = 2.15$, $\text{p}K_{\text{a}2} = 7.20$, $\text{p}K_{\text{a}3} = 12.32$.

pH =

Last name _____

3. Calculate the pH of the solution obtained by dissolving 5 mmol of PCl_5 in 1 L of a solution containing a stoichiometric amount of KOH.

pH =

The Young chemist had the following indicators on hand: methyl orange ($\text{pT} = 4.0$); methyl red ($\text{pT} = 5.5$) and phenolphthalein ($\text{pT} = 9.0$).

4. Calculate what the titration products will be when using different indicators. The possible outcomes are

a) $\text{KCl} + \text{H}_3\text{PO}_4$; б) $\text{KCl} + \text{KH}_2\text{PO}_4$; в) $\text{KCl} + \text{K}_2\text{HPO}_4$; г) $\text{KCl} + \text{K}_3\text{PO}_4$;

Fill in the table below and indicate the corresponding titration errors. Confirm with calculations after the table.

Indicator	Products	Error Δ , %
Methyl Orange		
Methyl Red		
Phenolphthalein		

Assume that the young chemist titrated a 10 mL aliquot of the solution from part 2 with a 0.03 M KOH solution.

Last name _____

Space for calculations

A 10 mL aliquot of the solution obtained after dissolving the sample from the jar was titrated by the young chemist with 0.03 M KOH solution. For the titration yielding $\text{KCl} + \text{KH}_2\text{PO}_4$ he used 6.37 mL of KOH solution. For the titration yielding $\text{KCl} + \text{K}_2\text{HPO}_4$, 7.54 mL was consumed.

Last name _____

5. Calculate what fraction of PCl_5 had hydrolyzed

$\alpha =$

6. What methods could you propose for the simultaneous determination of the amounts of phosphoric acid and hydrochloric acid?

Last name _____

Problem 6. Kinetics of Oxidation of Alcohols and Thiols by Dichromate (7 points)

Question	1	2	3	4	5	6	Total
Points	6	6	6	6	11	3	38
Score							

The oxidation of methanol by pyridinium dichromate in DMSO in acidic medium (in the presence of toluenesulfonic acid) proceeds according to the equation:



The kinetics of this reaction were studied by spectrophotometrically measuring the concentration of dichromate over time under conditions of large excess of the latter, at various concentrations of alcohol and acid.

In one experiment, with initial concentrations of dichromate $[(\text{PyH})_2\text{Cr}_2\text{O}_7]_0 = 0.001 \text{ M}$, alcohol $[\text{CH}_3\text{OH}]_0 = 0.05 \text{ M}$, and acid $[\text{TsOH}]_0 = 0.05 \text{ M}$, the dichromate concentration decreased over time as follows:

$t, \text{ min}$	0	10	25	50	80	100	150
$[(\text{PyH})_2\text{Cr}_2\text{O}_7] \cdot 10^3, \text{ M}$	1.00	0.93	0.83	0.69	0.55	0.47	0.32

1. Determine the reaction order with respect to dichromate and calculate the value of the effective rate constant under the given conditions. Show your calculations.

Order = _____ $k_{\text{eff}} =$ _____

Last name _____

2. Analogous experiments carried out at other concentrations of alcohol and acid gave the results shown below. Assuming that the reaction order with respect to dichromate in these experiments is the same as in question 1, calculate the effective rate constant for each experiment. Show your calculations..

[CH ₃ OH] ₀ = 0.1 M, [TsOH] ₀ = 0.05 M							
t, min	0	10	20	30	40	50	60
[(PyH) ₂ Cr ₂ O ₇]·10 ³ , M	1.00	0.86	0.74	0.64	0.55	0.47	0.41
$k_{\text{eff}} =$ _____							
[CH ₃ OH] ₀ = 0.05 M, [TsOH] ₀ = 0.1 M							
t, min	0	5	15	25	35	50	70
[(PyH) ₂ Cr ₂ O ₇]·10 ³ , M	1.00	0.93	0.80	0.69	0.59	0.47	0.35
$k_{\text{eff}} =$ _____							
[CH ₃ OH] ₀ = 0.15 M, [TsOH] ₀ = 0.1 M							
t, min	0	5	10	15	20	25	30
[(PyH) ₂ Cr ₂ O ₇]·10 ³ , M	1.00	0.80	0.64	0.51	0.41	0.32	0.26
$k_{\text{eff}} =$ _____							

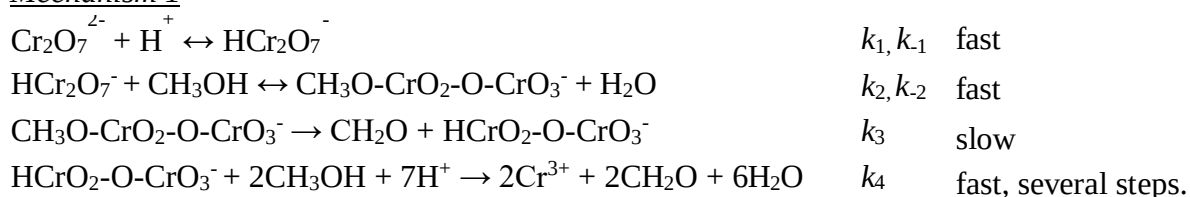
Last name _____

3. Using the results from questions 1 and 2, determine the reaction orders with respect to alcohol and acid, and the rate constant of the reaction. Show your calculations.

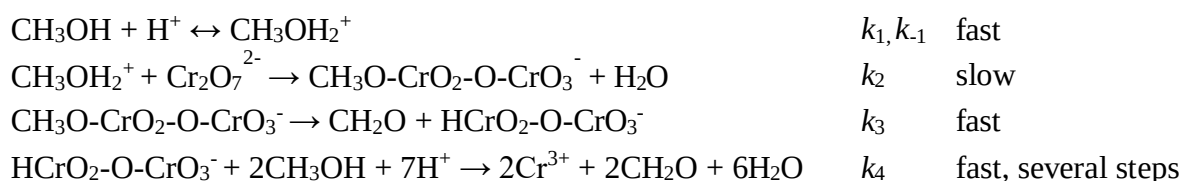
Order of CH₃OH = _____ Order of TsOH = _____ k_{eff} = _____

Two mechanisms have been proposed for this reaction:

Mechanism 1



Mechanism 2



4. For both proposed mechanisms, express the rate of formaldehyde formation in terms of reagent concentrations. Use reasonable approximations where necessary. Determine whether the kinetic equations you derived are consistent with the orders determined in question 3 (check the appropriate box).

Last name _____

Mechanism 1

Consistent with Question 3:

Y

☐

N

☐

Mechanism 2

Consistent with Question 3:

Y

☐

N

☐

Interestingly, replacing the oxygen atom with a sulfur atom in methanol significantly affects both the kinetic behavior of the reaction and its chemical outcome: the oxidation of methanethiol by pyridinium dichromate in acidic medium in the same solvent leads to the formation of dimethyl disulfide:



The reaction is first order with respect to dichromate:

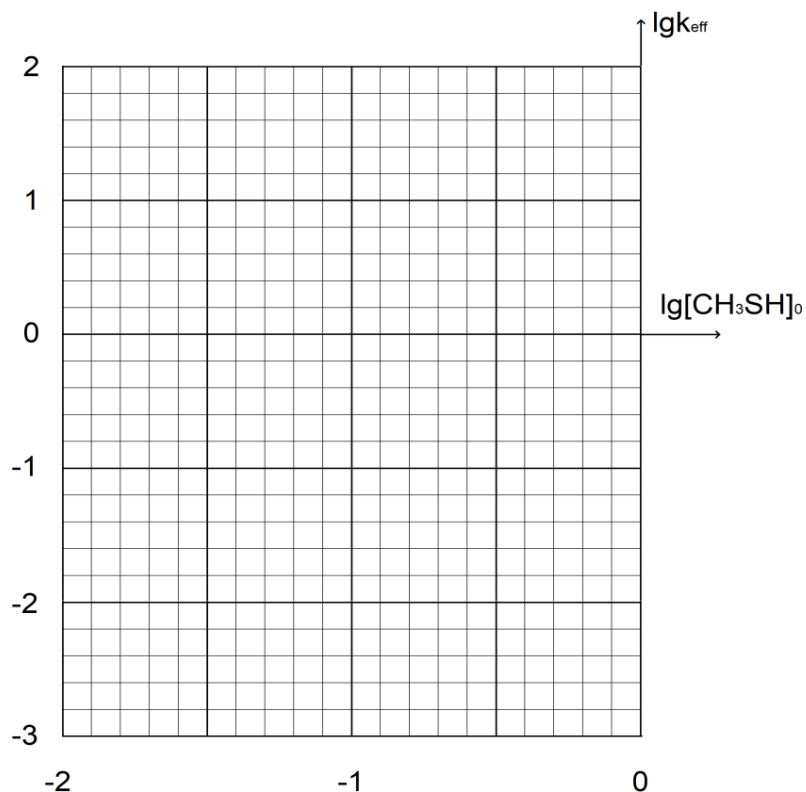
$$r = k_{\text{eff}}[(\text{PyH})_2\text{Cr}_2\text{O}_7]$$

Last name _____

The effective rate constant k_{eff} , was determined in 4 series of experiments at different initial concentrations of thiol and acid. In each individual series, the initial concentration of thiol was varied at a fixed initial concentration of acid. The values of k_{eff} , determined in this way are given in the table below.

Series 1: $[\text{TsOH}]_0 = 0.05 \text{ M}$				
$[\text{CH}_3\text{SH}]_0, \text{ M}$	0.03	0.05	0.10	0.25
$k_{\text{eff}}, \text{ min}^{-1}$	0.0031	0.0085	0.0340	0.213
Series 2: $[\text{TsOH}]_0 = 0.09 \text{ M}$				
$[\text{CH}_3\text{SH}]_0, \text{ M}$	0.03	0.05	0.10	0.25
$k_{\text{eff}}, \text{ min}^{-1}$	0.0099	0.0276	0.110	0.690
Series 3: $[\text{TsOH}]_0 = 0.23 \text{ M}$				
$[\text{CH}_3\text{SH}]_0, \text{ M}$	0.03	0.05	0.10	0.25
$k_{\text{eff}}, \text{ min}^{-1}$	0.0648	0.180	0.720	4.50
Series 4: $[\text{TsOH}]_0 = 0.40 \text{ M}$				
$[\text{CH}_3\text{SH}]_0, \text{ M}$	0.03	0.05	0.10	0.25
$k_{\text{eff}}, \text{ min}^{-1}$	0.196	0.545	2.18	13.6

5. Plot the measured values k_{eff} from the table on a graph with coordinates $\lg[\text{CH}_3\text{SH}]_0 - \lg k_{\text{eff}}$. Draw straight lines through the points corresponding to each individual species.



Last name _____

Determine the values of the quantities from the graph and enter them in the cells below:

Order with respect to thiol CH_3SH	
Y-intercept of the line for Series 1	
Y-intercept of the line for Series 2	
Y-intercept of the line for Series 3	
Y-intercept of the line for Series 4	

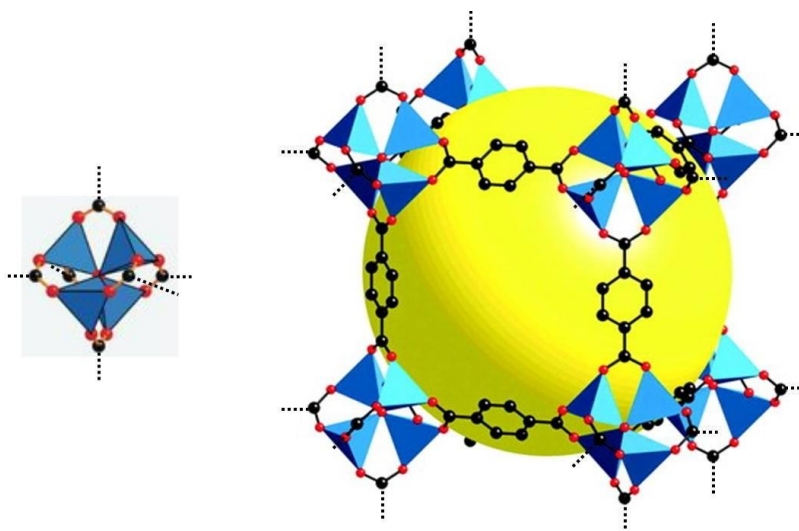
Calculate the reaction order with respect to acid TsOH and the rate constant of the reaction..

6. Propose a reaction mechanism consistent with the kinetic equation you found (without deriving it). Indicate the fast and slow steps. Take into account the following experimental fact: acrylonitrile added to the reaction mixture, purified of dissolved oxygen, does not polymerize.

Problem 7. “Guardians” and Conductors (7 points)

Question	1	2	3	4	5	6	7	8	9	10	Total
Points	1	1	2	2	1	2	4	2	3	2	20
Score											

Compounds whose structures contain large cavities or channels are extremely important as ionic conductors or "guardians" (storage materials). For instance, metal-organic framework structures (MOFs) are promising materials for hydrogen storage. The well-known compound MOF-5 consists of an inorganic building block containing zinc and an organic linker — the dianion of terephthalic acid. The zinc cations are in a tetrahedral environment of linker oxygens and an additional oxide ion, forming a Zn_4O^{6+} cation analogous to the cation in the structure of zinc(II) oxoacetate. The figure shows the structure of MOF-5 (right) and the inorganic block — zinc cations inside tetrahedra (left).



1. Determine the composition of MOF-5

Composition of MOF-5:

It is known that 1 g of MOF-5 can absorb 45 mg of hydrogen, which is accommodated in the large cavities shown as a sphere in the figure.

2. Determine the maximum number of hydrogen molecules per one formula unit of MOF-5.

Number of H_2 molecules per 1 formula unit of MOF-5: ____

Last name _____

Hydrogen can also be stored in inorganic structures; metallic palladium or intermetallics such as LaNi₅ are capable of dissolving it. Another intermetallic compound capable of storing hydrogen is FeTi, which crystallizes in the CsCl structure type with $a = 2.98 \text{ \AA}$. Hydrogen atoms are incorporated into the FeTi structure such that Ti and H together form an FCC sublattice, while the unit cell parameter remains practically unchanged.

3. Draw the unit cell of FeTiH_y.

4. Estimate the radius of hydrogen, taking the radius of titanium as $R(\text{Ti}) = 1.48 \text{ \AA}$.

$R(\text{H}) = \underline{\hspace{2cm}} \text{ \AA}.$

5. Estimate the degree of ionicity of the hydrogen-metal bond in FeTiH_y, given the covalent (0% ionicity) and ionic radii of hydrogen $R_{\text{cov}}(\text{H}) = 0.32 \text{ \AA}$, $R_{\text{ion}}(\text{H}) = 0.54 \text{ \AA}$.

Degree of ionicity: $\underline{\hspace{2cm}}\%$.

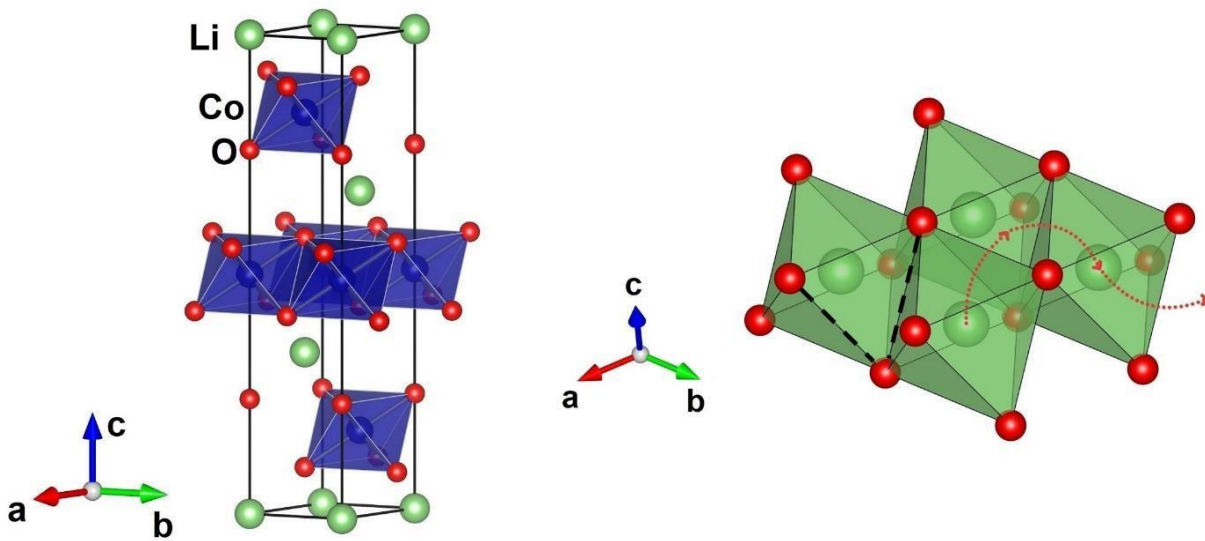
The actual occupancy of hydrogen sites in FeTi turns out to be no more than 2/3.

6. Compare MOF-5 and FeTi in terms of their hydrogen storage capacity by calculating the maximum mass fraction of H₂ in each.

$\omega(\text{H}_2 \text{ in MOF-5}) = \underline{\hspace{2cm}}\%, \quad \omega(\text{H}_2 \text{ in FeTi}) = \underline{\hspace{2cm}}\%.$

Last name _____

The presence of channels in structures can enable ionic transport. This is the basis for the search for cathode materials for metal-ion batteries. The crystal structure of the most well-known cathode material — LiCoO_2 — in lithium-ion batteries is shown in the figure below (hexagonal unit cell, $a = b = 2.81 \text{ \AA}$). It is known that the oxide ions form a 6-layer close-packed arrangement, and lithium and cobalt cations occupy all octahedral voids. The figure on the right shows the migration path of lithium cations — from an octahedral void first into a tetrahedral one, and then back into an octahedral one. The LiO_6 octahedra are slightly distorted such that the faces of the octahedron lying in the ab plane are regular triangles, while the remaining faces are isosceles triangles. The highlighted O–O distances are $3.09 \text{ \AA}$.

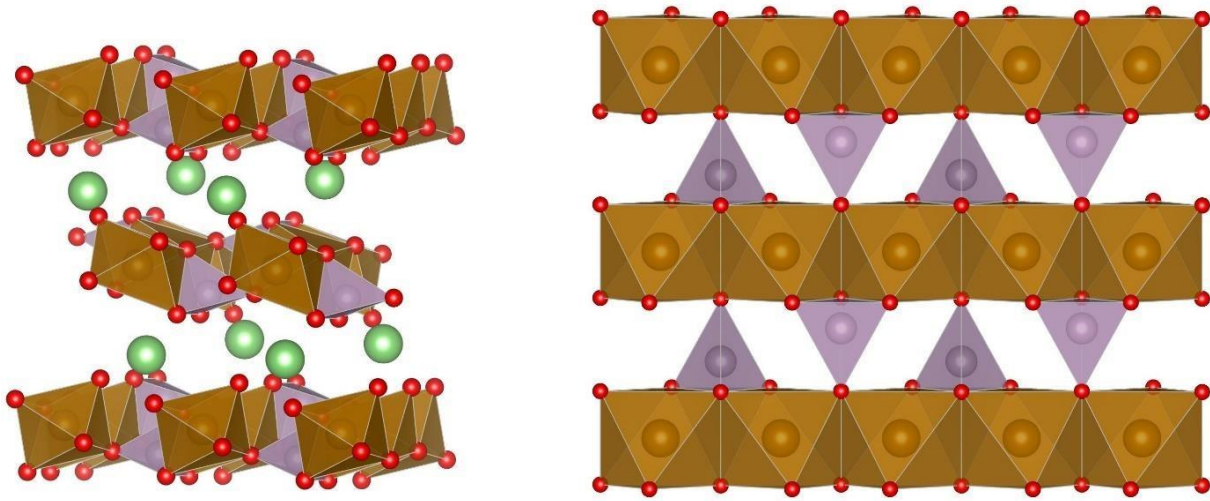


7. Calculate the maximum radius of a cation capable of passing through the face of the LiO_6 octahedron that forms an isosceles triangle. Take the ionic radius of oxygen as $R(\text{O}^{2-}) = 1.32 \text{ \AA}$.

$R_{\text{max}}(\text{Li}^+) = \underline{\hspace{2cm}} \text{ \AA}.$

Last name _____

Another cathode material is a complex phosphate of lithium and an unknown metal M^{2+} . The figure on the left shows the general view of the crystal structure. The M^{2+} cations and phosphorus form two-dimensional layers consisting of MO_6 octahedra and PO_4 tetrahedra. The right figure shows a top view of such a layer. Lithium cations are located between the layers.



8. Determine the composition of the layer and its charge. Identify the composition of the complex phosphate..

Layer Composition: _____

Complex phosphate composition: _____

It is known that this complex phosphate has an orthorhombic unit cell. The table gives the positions of the first three reflections in the diffraction pattern of the compound and their corresponding Miller indices.

$2\theta, ^\circ$	hkl
17.18	200
20.8	101
22.7	210

9. Calculate the unit cell parameters of the phosphate. The diffraction pattern was recorded using copper radiation with $\lambda = 1.5406 \text{ \AA}$. Use the formula:

$$\frac{1}{d^2} = \frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}$$

Last name _____

Fill in the table:

$2\theta, ^\circ$	$d, \text{\AA}$	hkl
17.18		200
20.8		101
22.7		210

$a =$ _____ $\text{\AA}$, $b =$ _____ $\text{\AA}$, $c =$ _____ $\text{\AA}$.

The density of the cathode material in the charged state was $d = 3.61 \text{ g}\cdot\text{cm}^{-3}$.

10. Identify the unknown metal M.

$M(M) =$ _____ $\text{g}\cdot\text{mol}^{-1}$

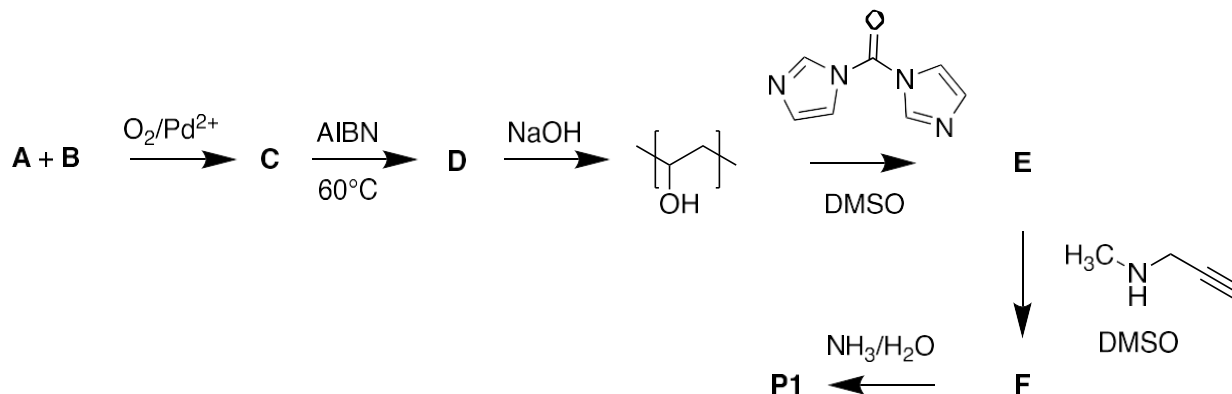
M – _____

Last name _____

Problem 8. Polymers (7 points)

Question	1	2	3	4a	4b	5a	5b	Total
Points	7	4	1	2	3	2	1	26
Score								

Polymer **P1** was obtained according to the following scheme:



AIBN = 2,2'-azobisisobutyronitrile. Before obtaining **E**, polyvinyl alcohol was fractionated to isolate a polymer with a number-average molecular weight of 15860. For the reaction starting from 1.00 g of polyvinyl alcohol, 1.84 g of 1,1'-carbonyldiimidazole and 43.6 mg of propargylmethylamine were used. Assume that the reactions at the steps of obtaining **E**, **F**, and **P1** are quantitative. In calculations, neglect the mass of macromolecular end groups.

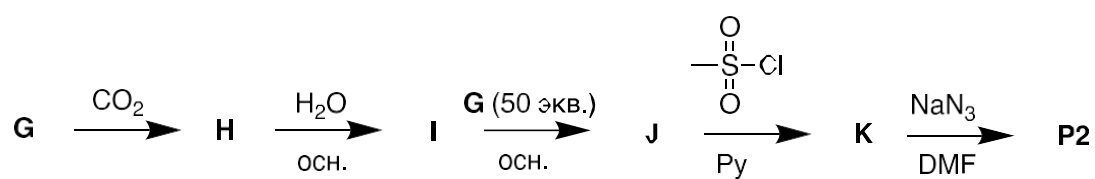
1. Draw the structures of **A–F** and **P1**. For polymers, provide the repeating unit structure; for copolymers, provide the number of each type of unit. Confirm numerical values with calculations.

Calculations:

Last name _____

A	B
C	D
E	
F	
P1	

Polymer **P2** was obtained according to the following scheme:



Last name _____

The transformation **G**→**H** is catalytic; Py = pyridine; base = base catalysis; the molecular formula of **G** coincides with that of one of the repeating units of **P1**.

2. Draw the structures of **G–K** and **P2**. For polymers, provide the repeating unit structure; for copolymers, provide the number of each type of unit. Confirm numerical values with calculations.

Calculations:	
G	H
I	J
K	P2

Last name _____

Polymer **P3** was obtained analogously to **P2**, but using methanol instead of reagent **I**.

3. Draw the structure of **P3** taking into account end groups and degree of polymerization.

P3

4. To 1.000 g of **P1**, polymer **P2** was added such that the amounts of the reacting functional groups in the resulting mixture were equal, along with an additional 65 mg of **P3**.
- Calculate the mass of **P2**.
 - Calculate the number-average molecular weight of the product at 50% conversion in the click reaction (consider conversion based on the functional groups present in the mixture in deficit).
 - Calculate the conversion in the click reaction corresponding to gel formation (a polymer with an infinitely large number-average molecular weight).

When answering this question, it is not allowed to use ready-made (i.e., without derivation) equations that relate conversion, molecular weight, and mixture composition.

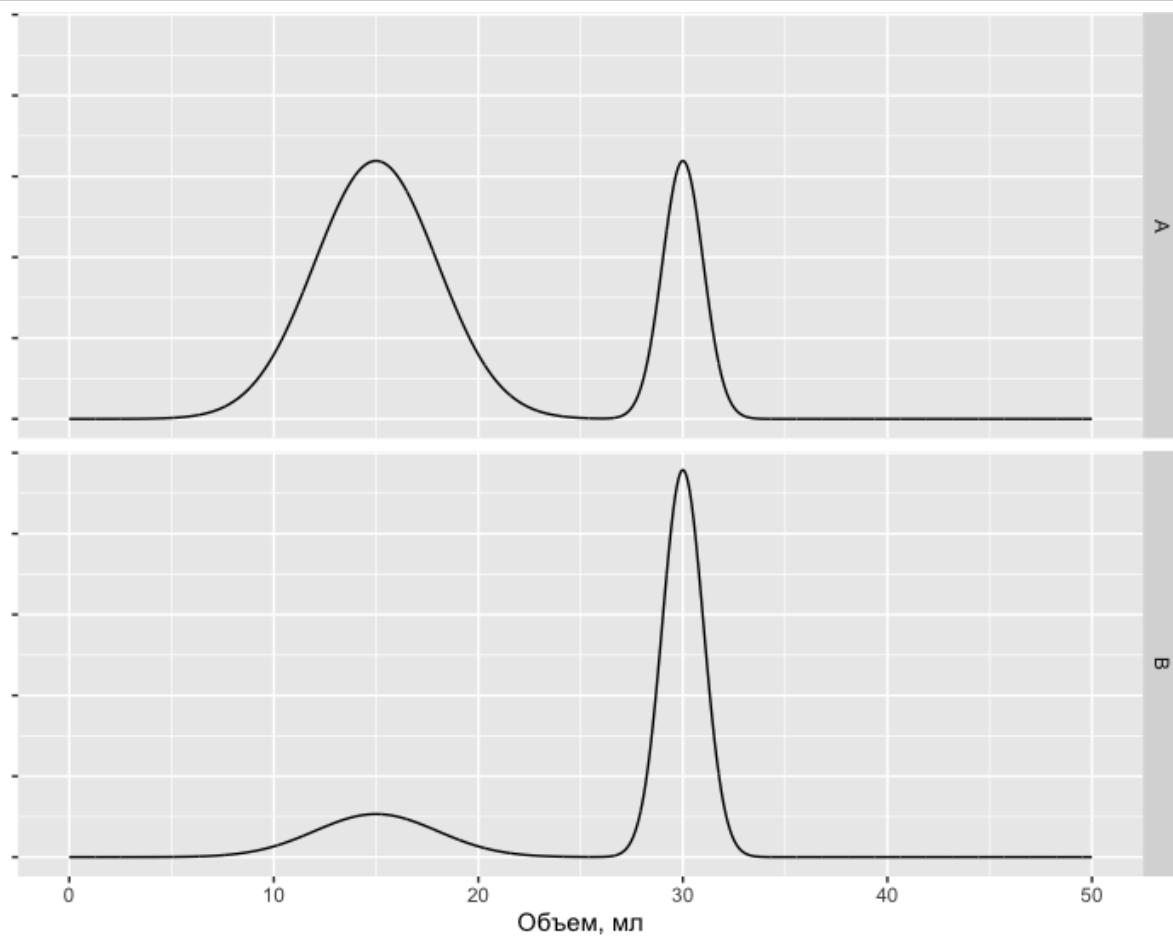
Last name _____

5. The answer field shows the results of gel permeation chromatography (GPC) for mixtures (A and B) of polymers **P1** and **P3**, whose molar ratios of reacting functional groups are 1:5 and 5:1.

a) Match the chromatograms with the composition of the mixtures.

b) On the same figure, depict the chromatograms of the products after the completion of the "click" reaction.

If necessary, briefly explain the answer in point b). The choice in point a) is evaluated only if an explanation is provided.

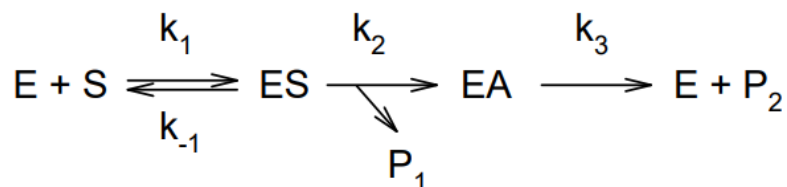


Last name _____

Problem 9. Three-Stage Scheme of Enzymatic Reactions (7 points)

Question	1	2	3	4	5	6	7	8	9	Total
Points	2.5	2.5	1	0.5	1	0.5	1	0.5	0.5	10
Score										

Many proteolytic enzymes catalyze reactions according to a three-stage scheme (EA – acyl-enzyme)



Assume that $[S]=[S]_0$ and $[E]_0 \ll [S]_0$, and that the standard form of the equation for the initial reaction rate is:

$$v_0 = \frac{k_{\text{cat,app}} [E]_0 [S]_0}{K_{\text{M,app}} + [S]_0}$$

A quasi-steady state with respect to ES and EA has been established in the system.

1. Derive an expression for the initial reaction rate $V_{01}=d[P_1]/dt$.

Derivation:

$k_{\text{cat,app}} =$ _____

$k_{\text{M,app}} =$ _____

Last name _____

2. Derive an expression for the initial reaction rate $V_{02} = d[P_2]/dt$.

Derivation:

$k_{cat,app} =$ _____

$k_{M,app} =$ _____

α -Chymotrypsin catalyzes the hydrolysis not only of peptide bonds, but also of amide and ester bonds; the rate-limiting steps of these processes may differ. Thus, in the hydrolysis of p-nitroanilide of N-benzoyl-L-tyrosine (BTNA) the rate-limiting step is acylation of the enzyme, whereas in the hydrolysis of ethyl ester of N-acetyl-L-tyrosine (ATEE) the rate-limiting step is deacylation of the acyl-enzyme.

3. Write the hydrolysis equations for a) BTNA, b) ATEE. Use structural formulas for substrates and products. Identify P_1 and P_2 in each case.

a)

b)

Last name _____

4. Which rate constant is smaller in each case?

a) BTNA _____	6) ATEE _____
---------------	---------------

5. Simplify $k_{\text{cat,app}}$ for the hydrolysis of BTNA and ATEE when measuring the reaction rate by P_1 and P_2 , if in each case k_2 and k_3 differ by more than 2 orders of magnitude.

BTNA, P_1 $k_{\text{cat,app}} =$	ATEE, P_1 $k_{\text{cat,app}} =$
BTNA, P_2 $k_{\text{cat,app}} =$	ATEE, P_2 $k_{\text{cat,app}} =$

6. What relationships between the rate constants must hold for the experimental data to be satisfactorily described by the quasi-equilibrium approximation when monitoring the reaction by P_1 ?

Answer: _____

An additional nucleophile Nu was added to the system, with rate constant k_4 for the step involving it.

7. Write the kinetic scheme for this case

8. How will the observed initial rate of the hydrolysis reaction change in this case?

a) BTNA _____	6) ATEE _____
---------------	---------------

9. The concentration(s) of which product(s) should be monitored for reliable determination of the initial reaction rate if Nu is present in the system?

Answer: _____